Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 3 · SLO 1 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 3 : SLO â€“ 1 | Advantages and Disadvantages of an Array

I. Advantages â€” Fill in the Blanks

1. direct (random/constant-time) â€” index-based O(1) access.
2. contiguous â€” elements sit in contiguous memory locations.
3. fixed (large) â€” good for storing a fixed number of similar-typed items.
4. fixed â€” size is fixed once declared.
5. repetitive (iterative) â€” loops make bulk operations easy.
6. overhead (need) â€” no need to declare many separate variables.
7. known (exact/sufficient) â€” memory is efficient when the size matches the need.
8. constant (O(1)) â€” index lookup time.

II. Disadvantages â€” Fill in the Blanks

1. compile (declaration) â€” size must be known at compile time (for static arrays).
2. fixed â€” size cannot change at runtime.
3. shifting â€” middle insertion requires shifting other elements.
4. shifting â€” deletion also involves shifting elements.
5. too large (over-declared) â€” over-declaring wastes memory.
6. same â€” only one data type per array.
7. O(n) (linear) â€” worst-case search time in an unsorted array.